

# Mobile Communication

# What is Mobile Communication?

"Ability to use technology to wirelessly connect to and use centrally located information and/or application software through small, portable, computing & communication devices"

- Offers mobility with computing power
- Facilitates a large number of applications on a single device such as voice communication, SMS, MMS, M-commerce, M-learning

# Why Mobile Communication ?

- People are mobile



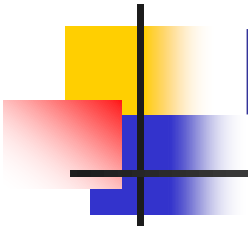
- Devices are mobile



# What is Mobility?



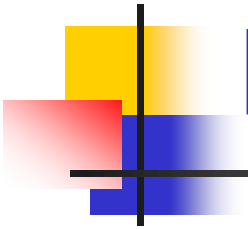

- **A person who moves**
  - Between different geographical locations
  - Between different networks
  - Between different communication devices
  - Between different applications
- **A device that moves**
  - Between different geographical locations
  - Between different networks



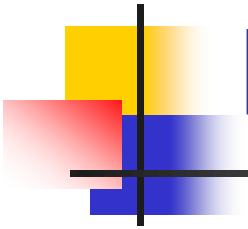
# Mobile Applications - 1

- Vehicles
  - transmission of news, road condition etc.
  - ad-hoc network with near vehicles to prevent accidents
- Emergencies
  - early transmission of patient data to the hospital
  - ad-hoc network in case of earthquakes, cyclones
  - military ...

# Mobile Applications - 2

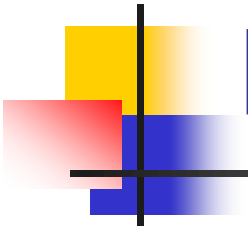


- Travelling salesmen
  - direct access to central customer files
  - consistent databases for all agents
  - mobile office
- Web access
  - outdoor Internet access
  - intelligent travel guide with up-to-date location dependent information



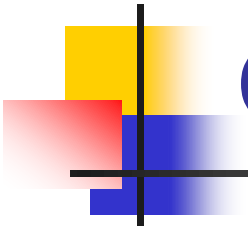
# Mobile Applications - 3

- Information services
  - push: e.g. stock quotes
  - pull: e.g. nearest cash ATM
- Entertainment
  - ad-hoc networks for multi user games
- others
  - mobile agents, e.g., shopping



# Mobile Devices

- Mobile Phone
  - Applications of mobile phones no longer confined to telephonic communication
  - Can synchronize and upload and download data to and from PCs
- Smart Phone
  - Functions as a phone as well as a PIM—phone diary, address book, task-list, calculator, alarm, and calendar
  - Integration of location information, GPS and maps
- Multimedia Phone
  - Enable users to enjoy mobile TV using EDGE/EGPRS (3G), 4G connectivity



# Operating System- **Symbian OS**

- widely used operating system for phones
- Runs exclusively on ARM processors
- Structure much like that of some desktop operating systems
- Initially designed for handheld devices with limited resources, strongly emphasizes on memory conservation
- Embodies event-based programming and when applications are not directly concerned with events, the CPU is switched off
- Such techniques are very useful in conserving battery life

# Operating System- **Windows CE**

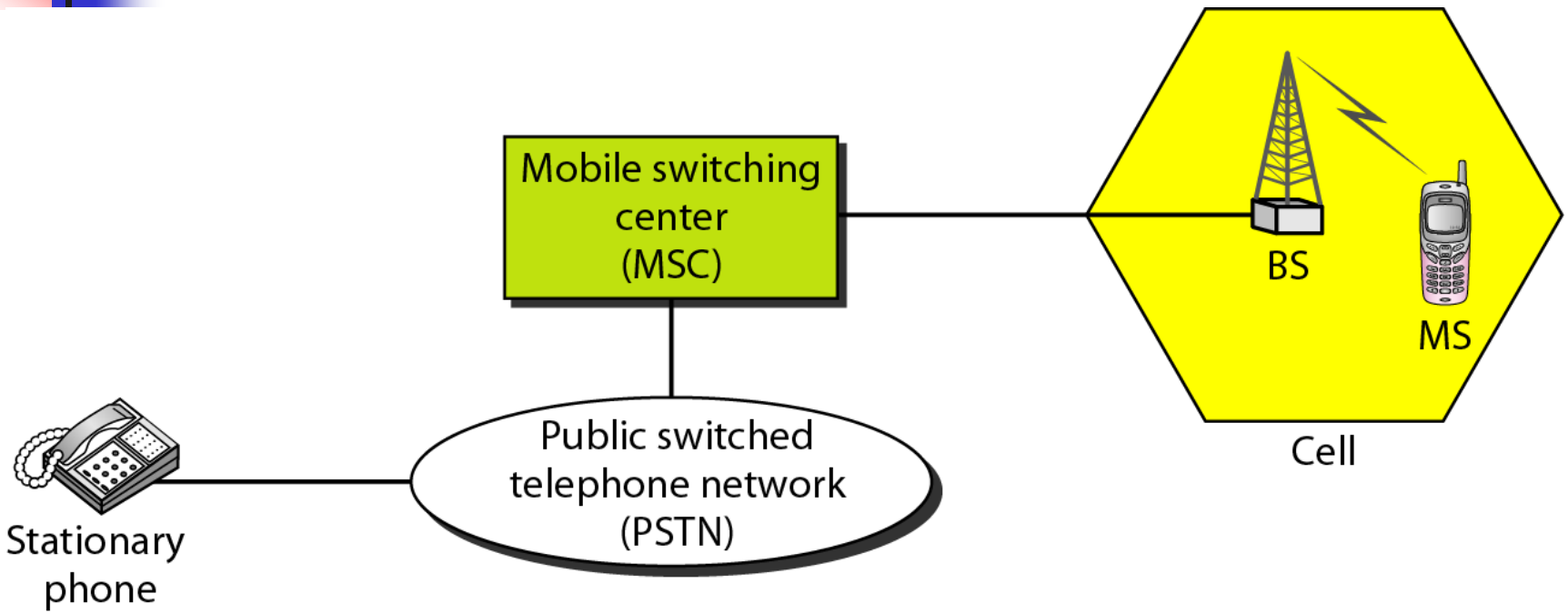
- An operating system from Microsoft
- Support multitasking on handheld devices
- Kernel different from the kernel of the desktop versions of Windows
- Offers PIM, MS Office, Internet Explorer etc.
- Windows CE OS memory needs are larger
- Support a wider range of hardware
- Support different CPUs such as NEC MIPS, Intel StrongARM, AMD X86, etc.

# Operating System- **Android**

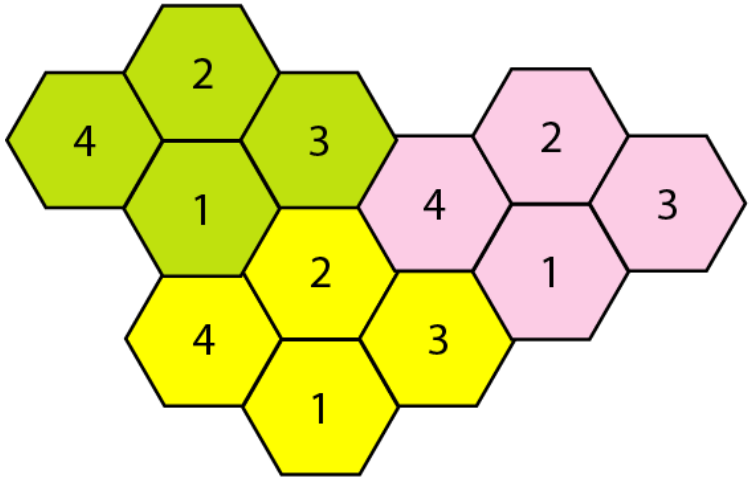
- initially developed by Android Inc., a firm purchased by Google in 2005.
- based upon a modified version of the Linux kernel.
- Live Folders for quick-viewing your data
- Video recording and sharing
- Faster, standards-compliant browser
- Many more features…

- Task - Explore other mobile OSs namely iOS

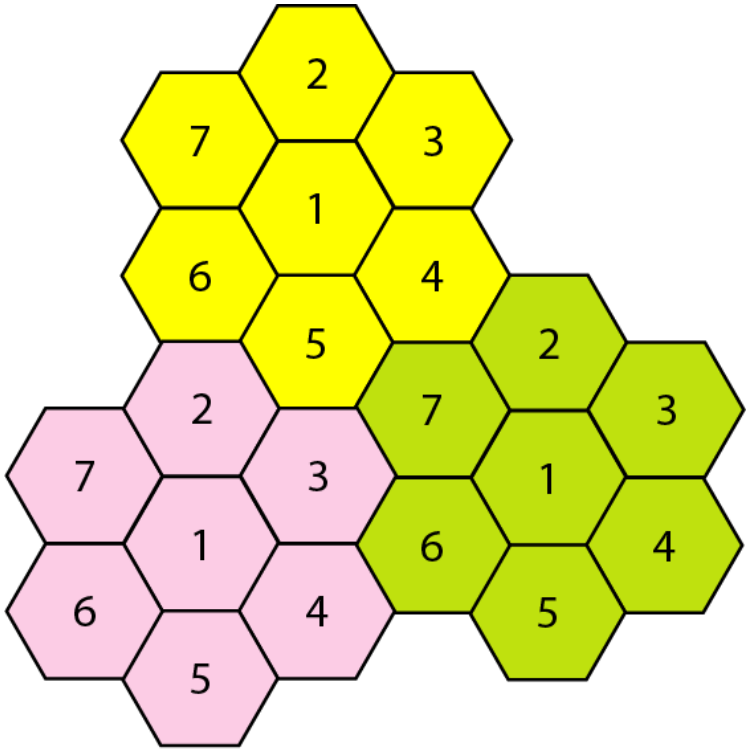
# *Cellular system*



# *Frequency reuse patterns*

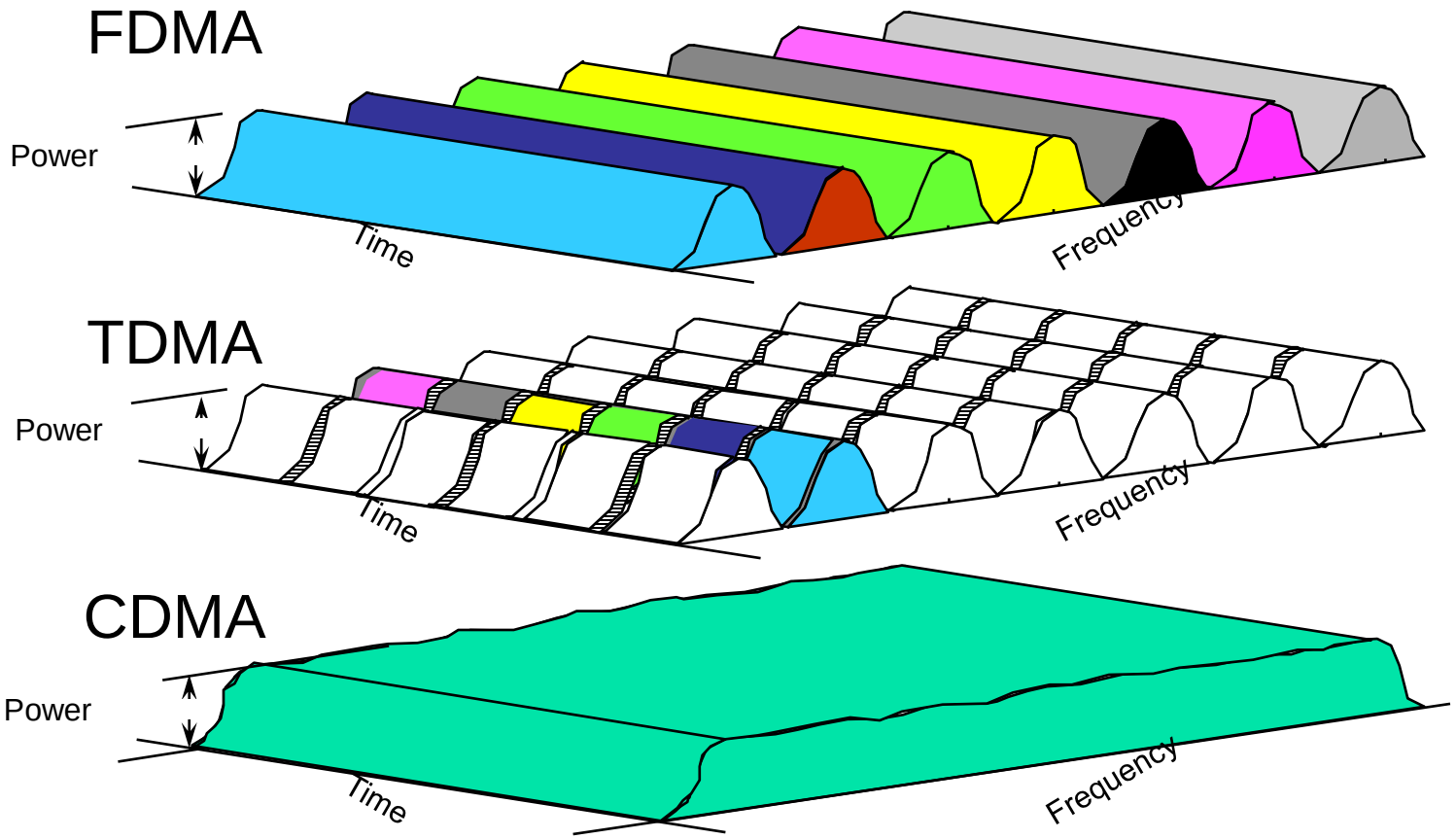



a. Reuse factor of 4

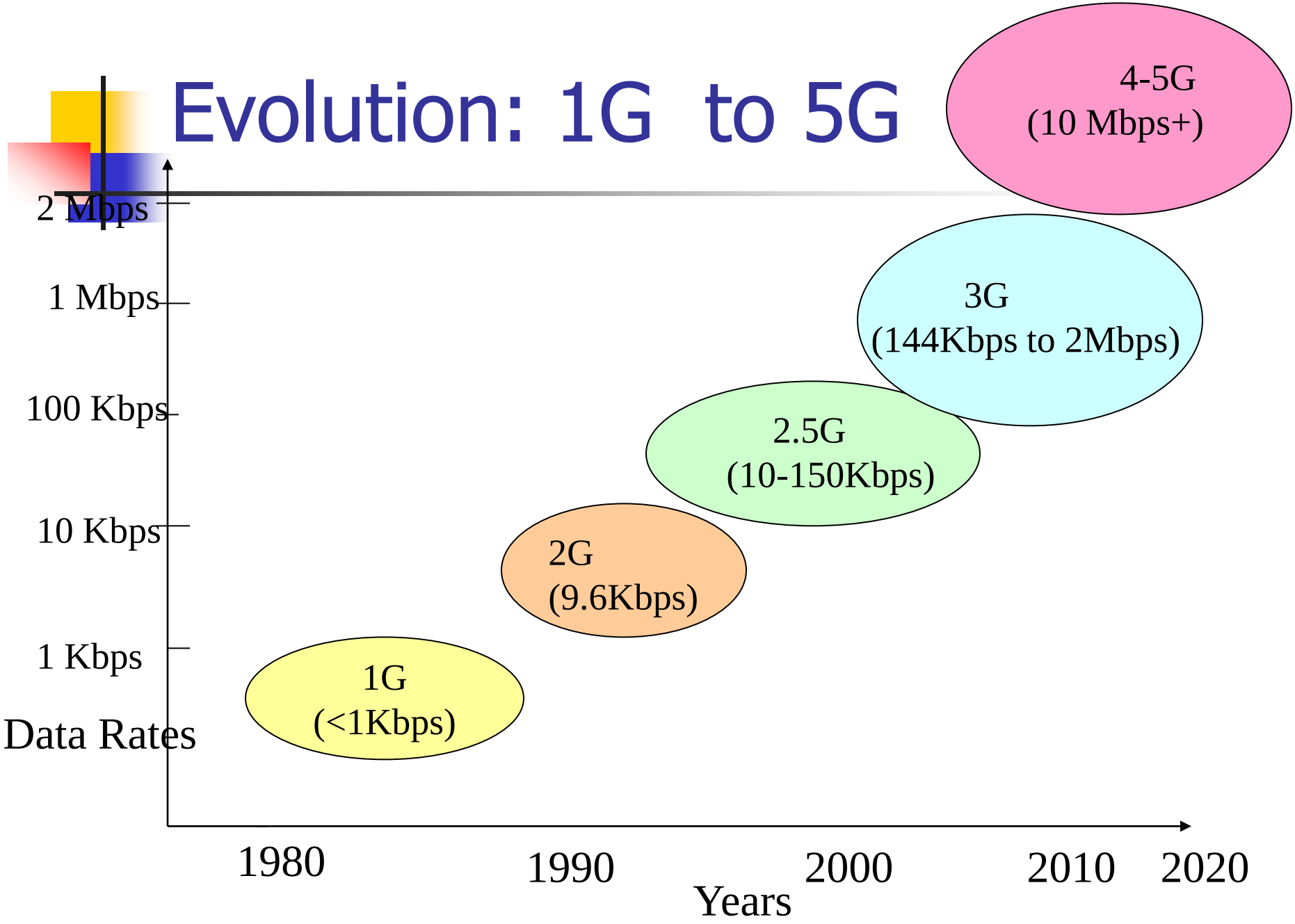



b. Reuse factor of 7

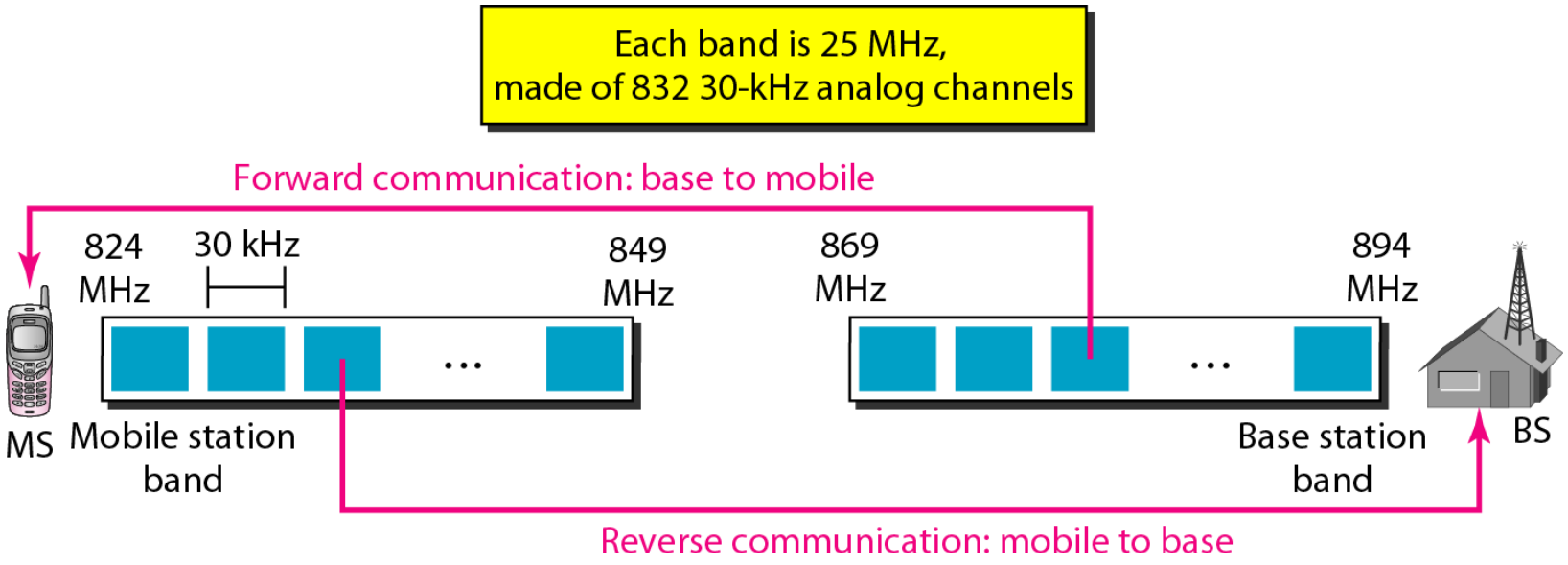
# Multiple Access Schemes



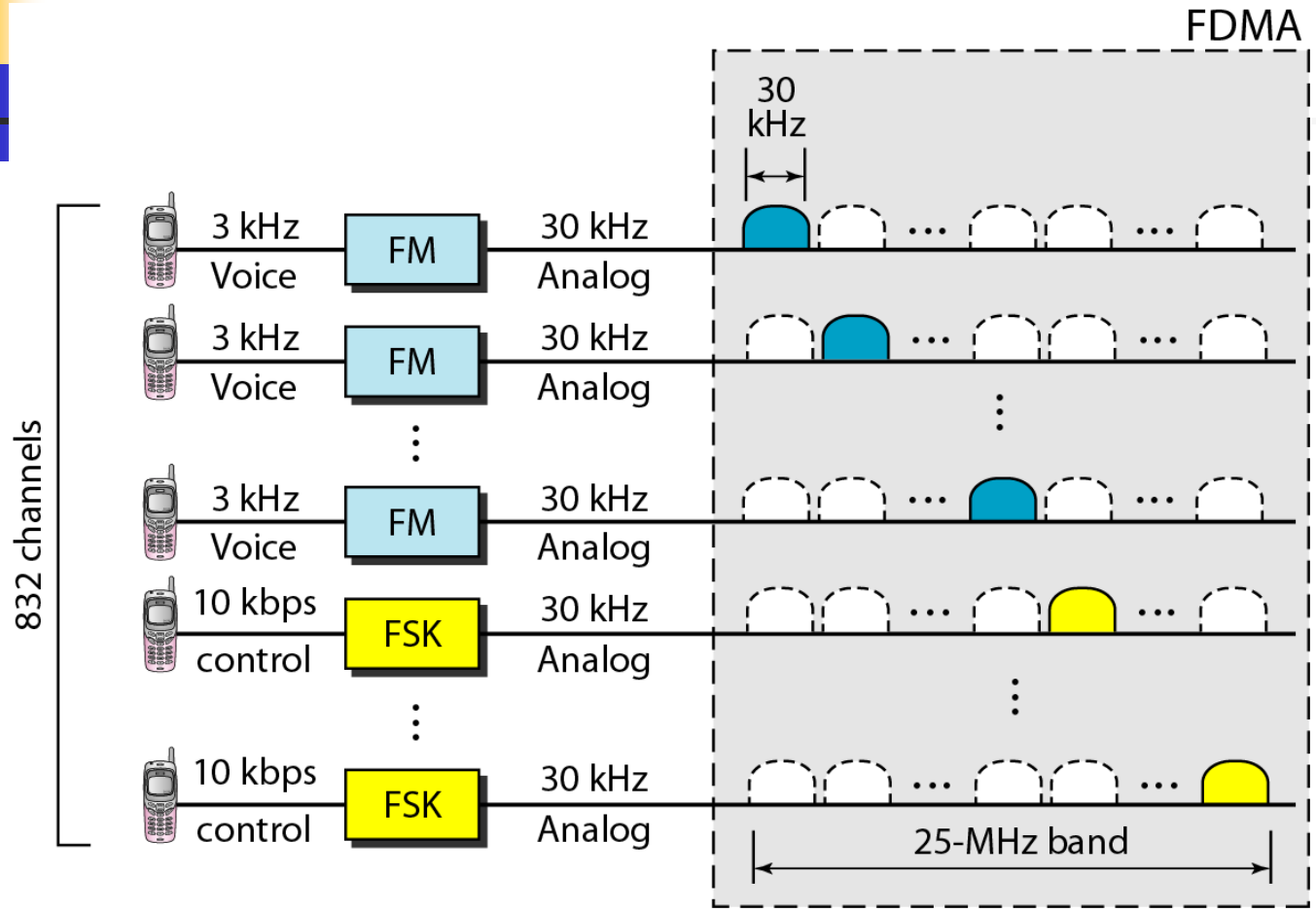
# Evolution: 1G to 5G



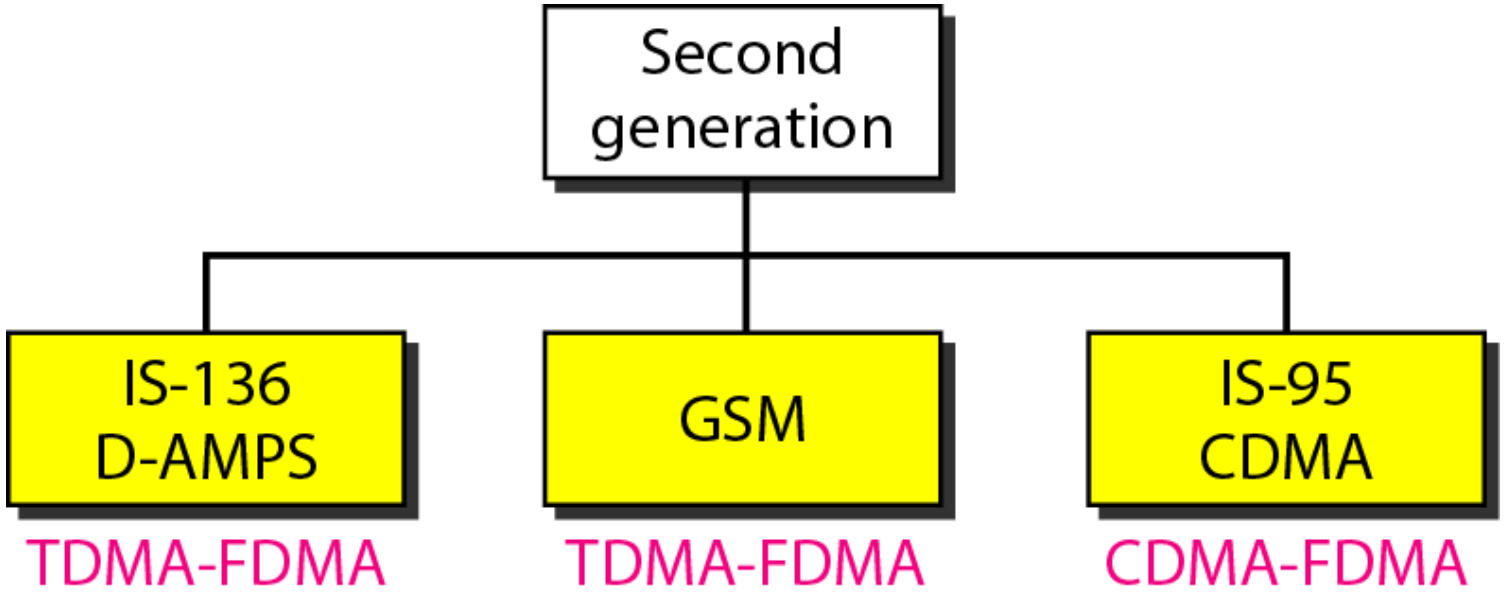
## Analog Mobile Phone System using FDMA



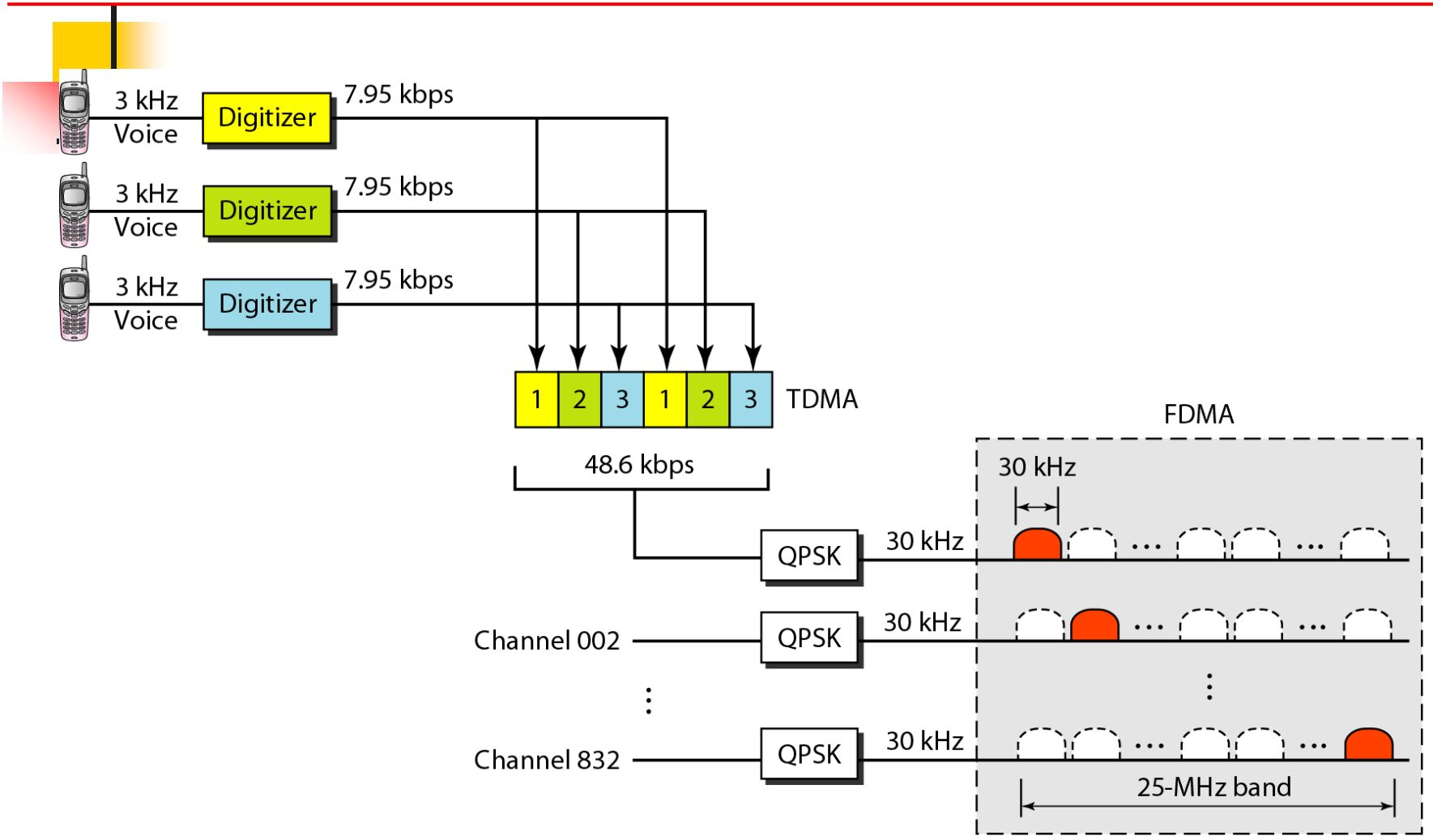
# *AMPS reverse communication band*



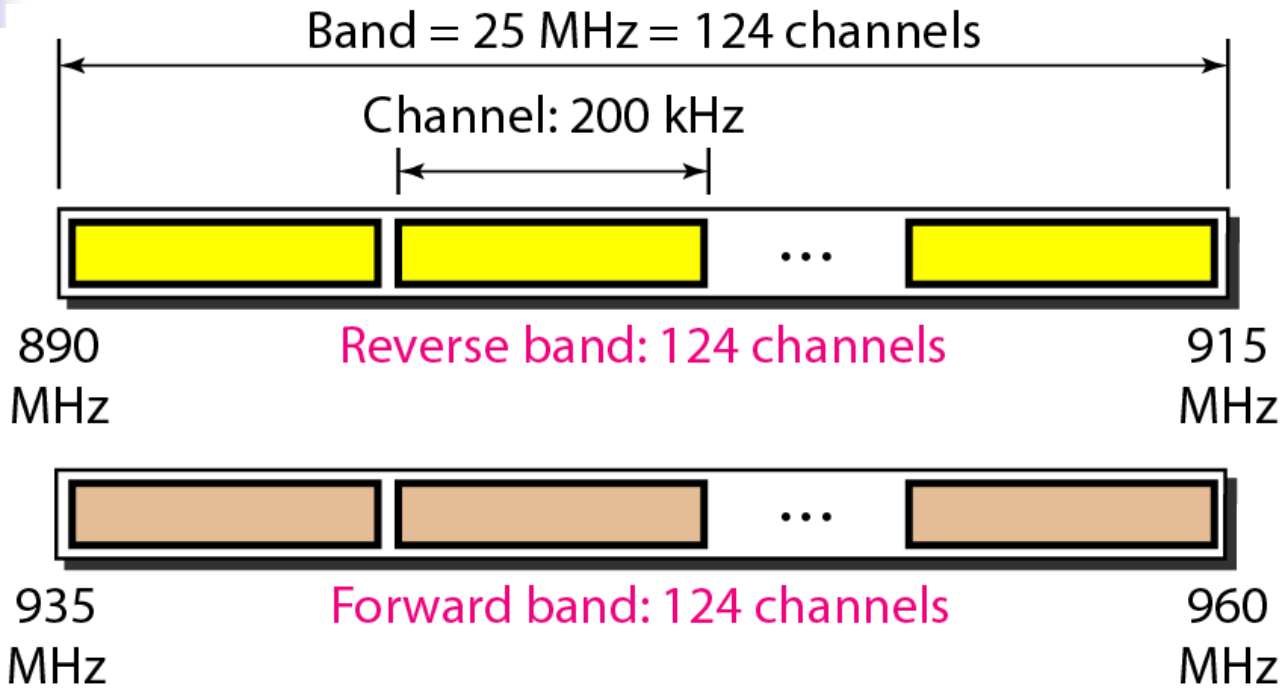
# 2G


Second generation
IS-136 D-AMPS
GSM
IS-95 CDMA
TDMA-FDMA
TDMA-FDMA
CDMA-FDMA

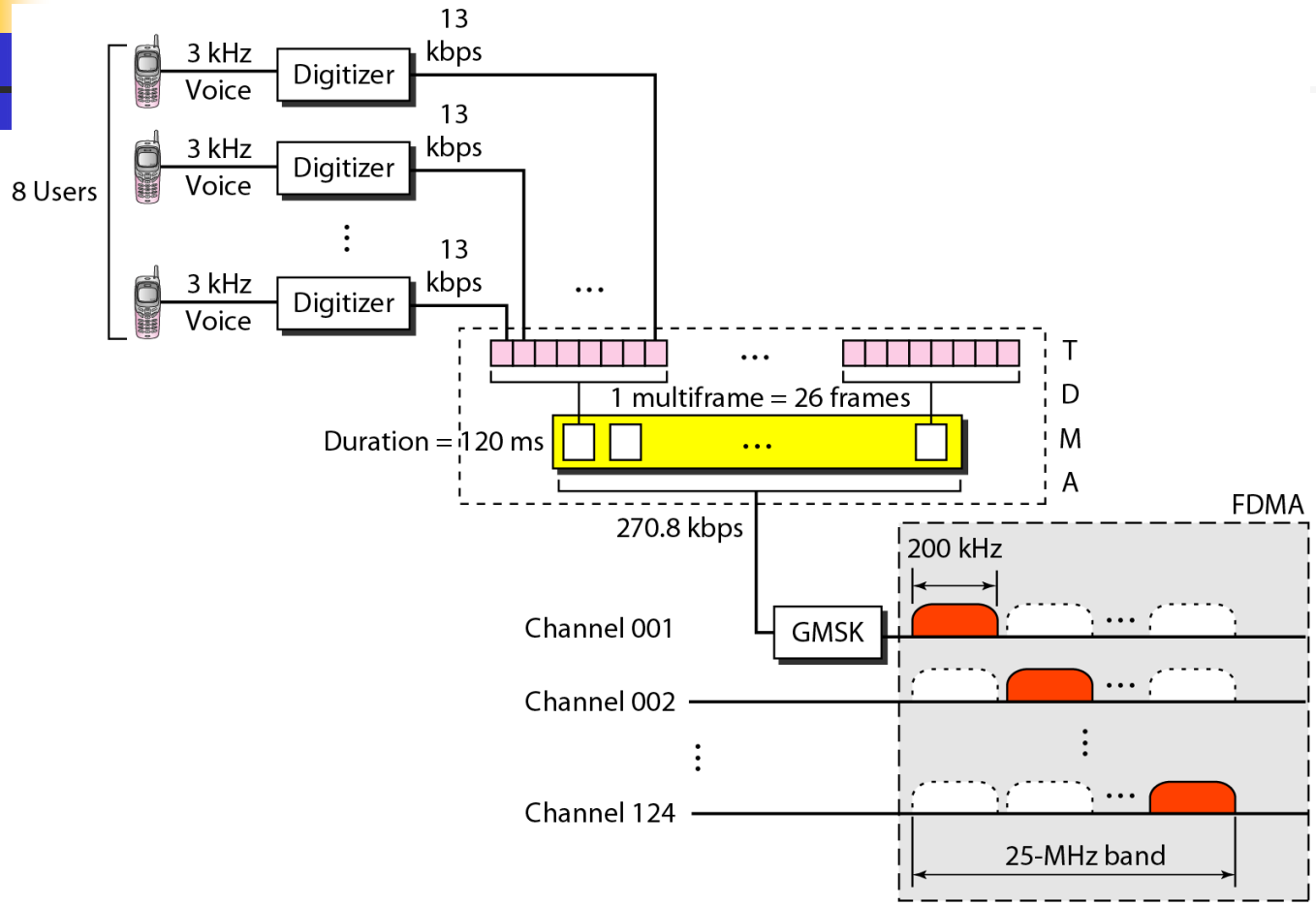
# D-AMPS


3 kHz
Voice
Digitizer
7.95 kbps
3 kHz
Voice
Digitizer
7.95 kbps
3 kHz
Voice
Digitizer
7.95 kbps
1
2
3
1
2
3
TDMA
48.6 kbps
FDMA
30 kHz
QPSK
30 kHz
Channel 002
QPSK
30 kHz
Channel 832
QPSK
30 kHz
25-MHz band

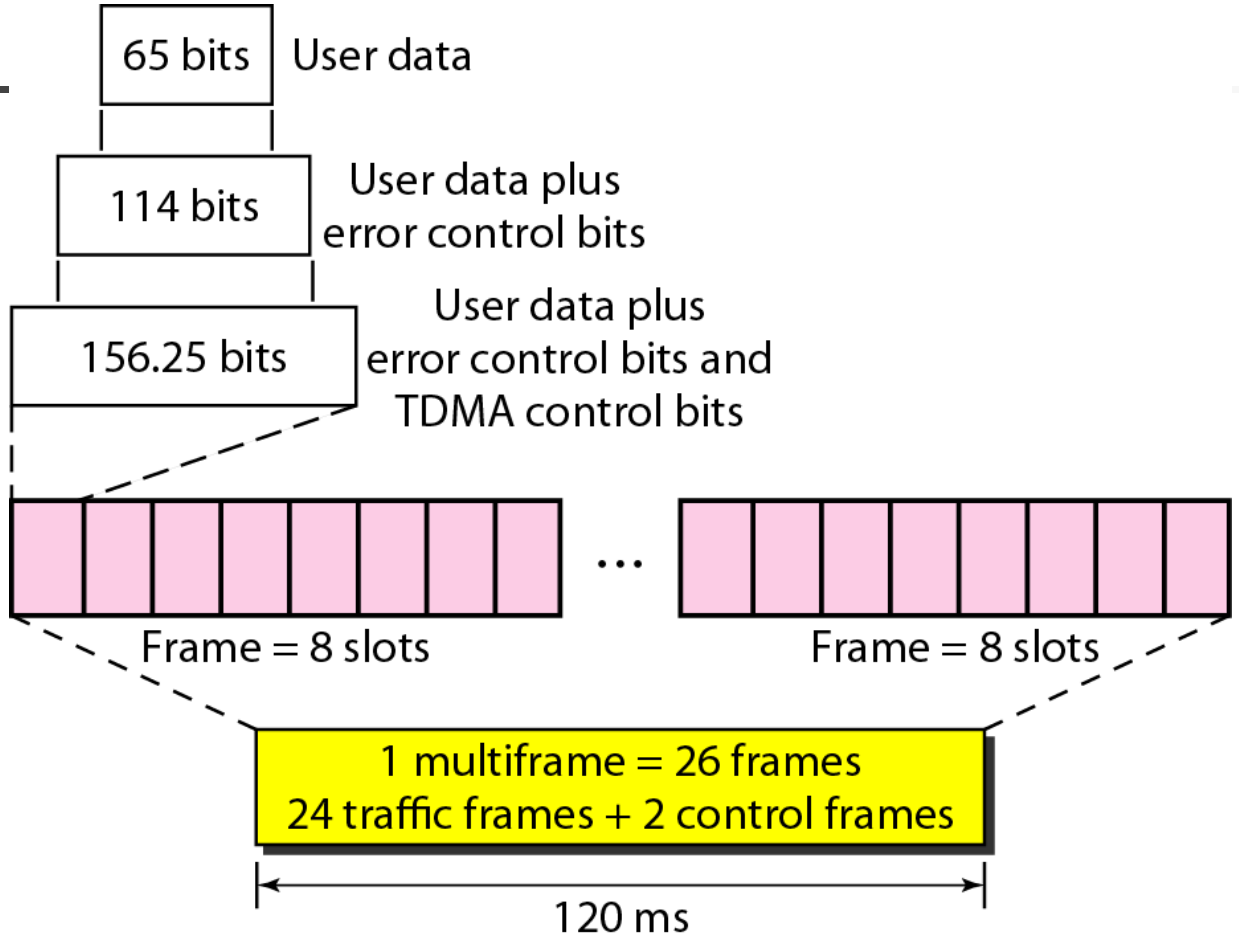
*GSM bands*


Band = 25 MHz = 124 channels
Channel: 200 kHz
890 MHz
Reverse band: 124 channels
915 MHz
935 MHz
Forward band: 124 channels
960 MHz

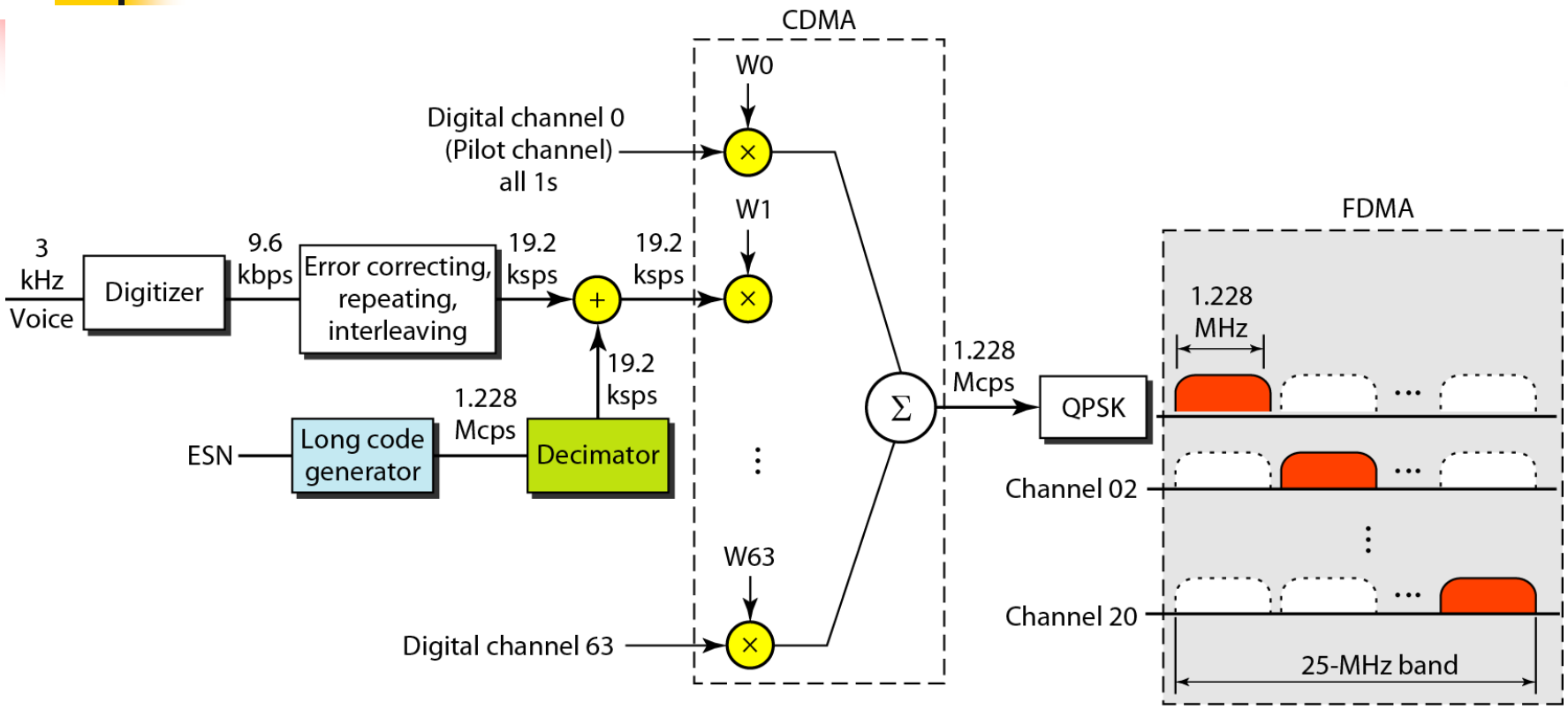
# GSM


8 Users
3 kHz Voice
Digitizer
13 kbps
3 kHz Voice
Digitizer
13 kbps
3 kHz Voice
Digitizer
13 kbps
1 multiframe = 26 frames
Duration = 120 ms
TDMA
270.8 kbps
FDMA
200 kHz
GMSK
Channel 001
Channel 002
Channel 124
25-MHz band

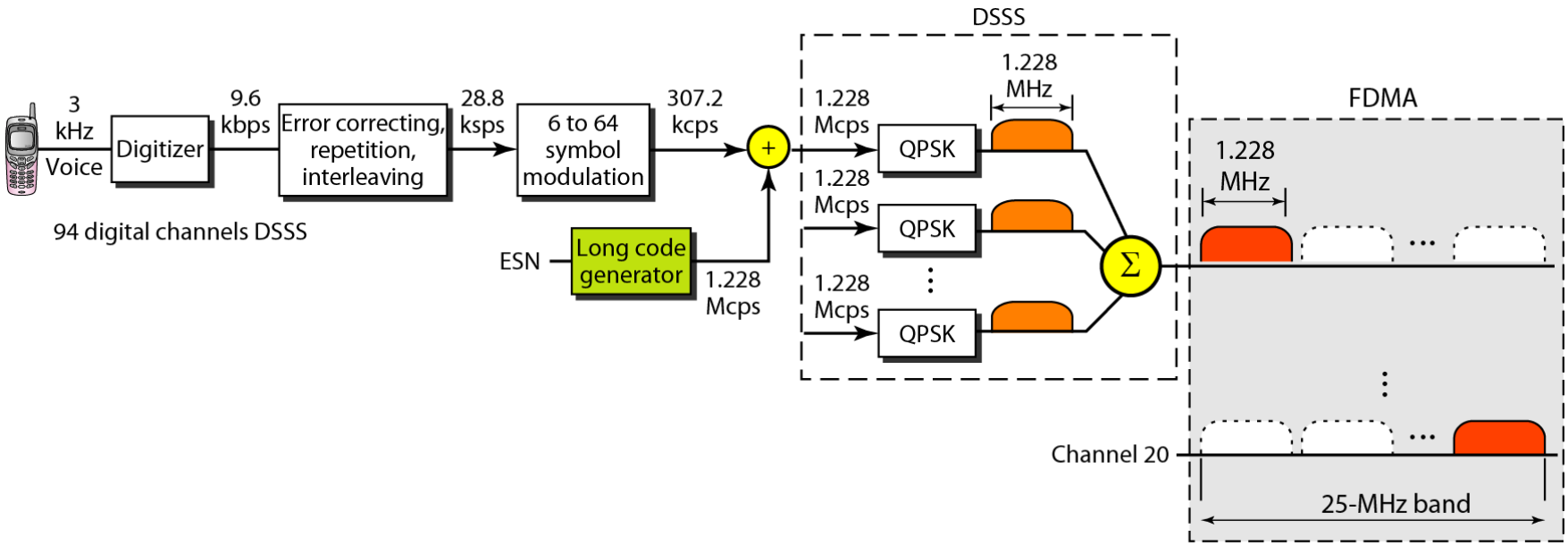
# *Multiframe components*



# *IS-95 forward transmission*



# *IS-95 reverse transmission*

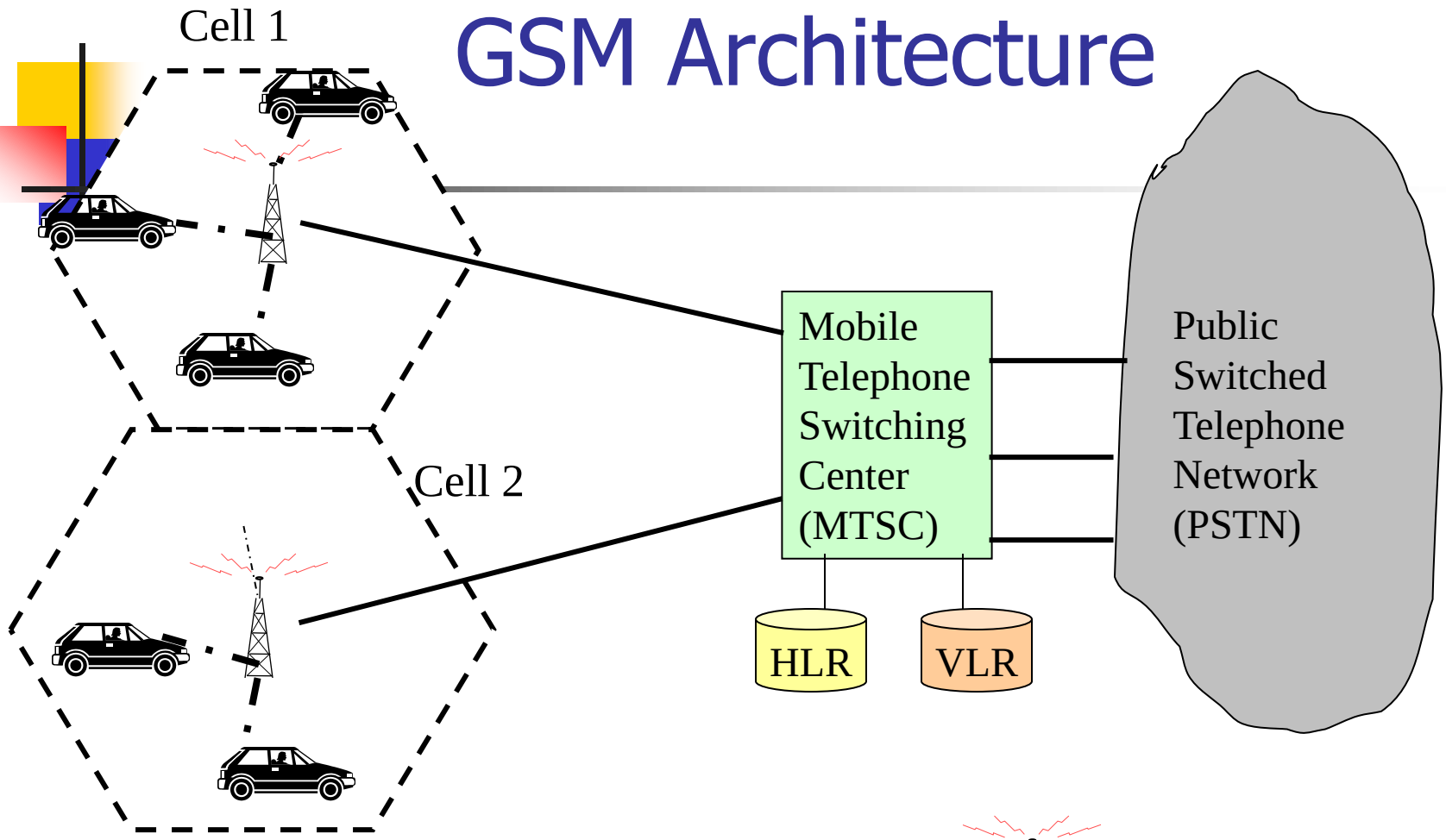




# GSM

- GSM
  - formerly: Groupe Spéciale Mobile (1982)
  - now: Global System for Mobile communication
- Communication: voice and data services
  - Telephony
  - Short Messaging Service (SMS)
  - Call forwarding, conferencing

# GSM Architecture




Mobile User

Cordless connection

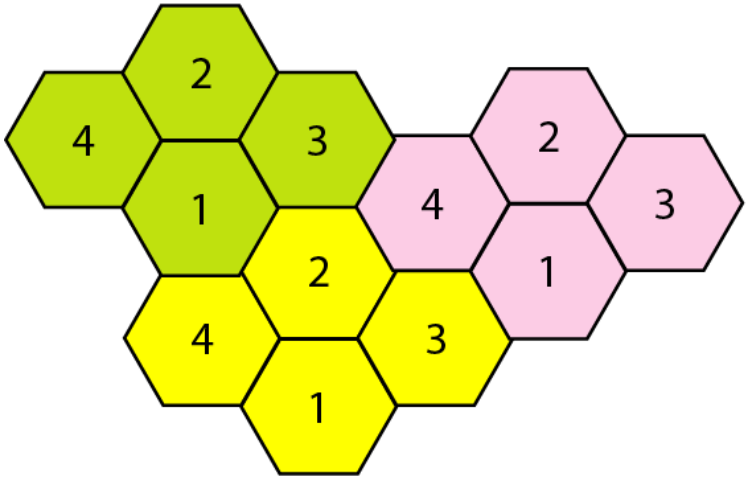
Wired connection

Base Transceiver Station (BTS)

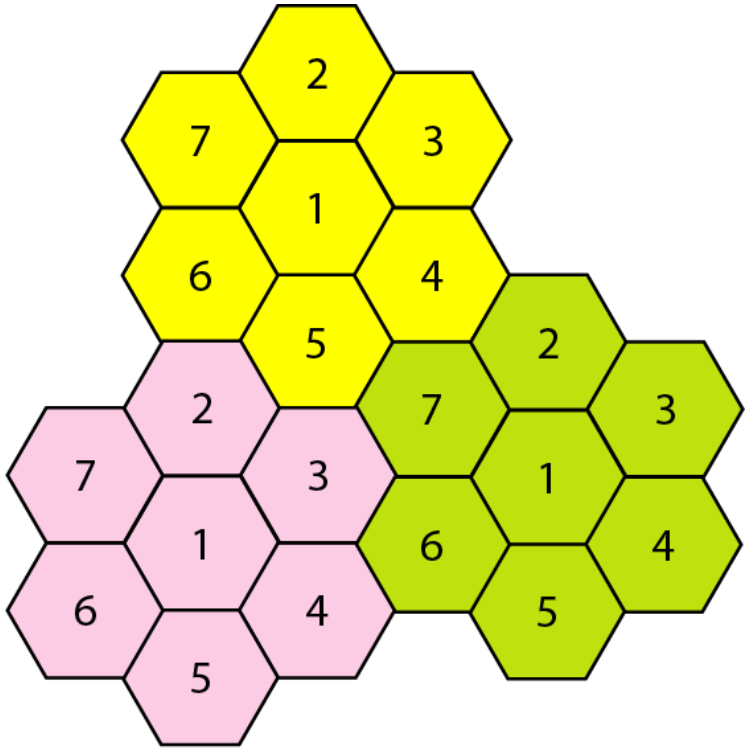
HLR = Home Location Register

VLR = Visitor Location Register

# Frequency Reuse



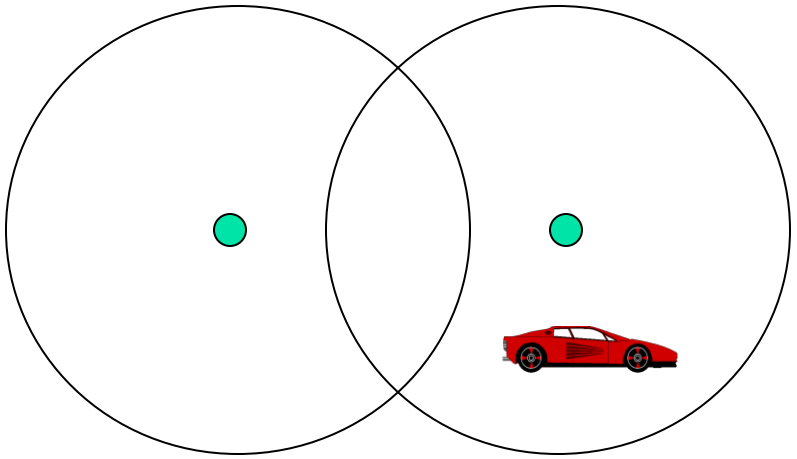

a. Reuse factor of 4




b. Reuse factor of 7

# Hand-off

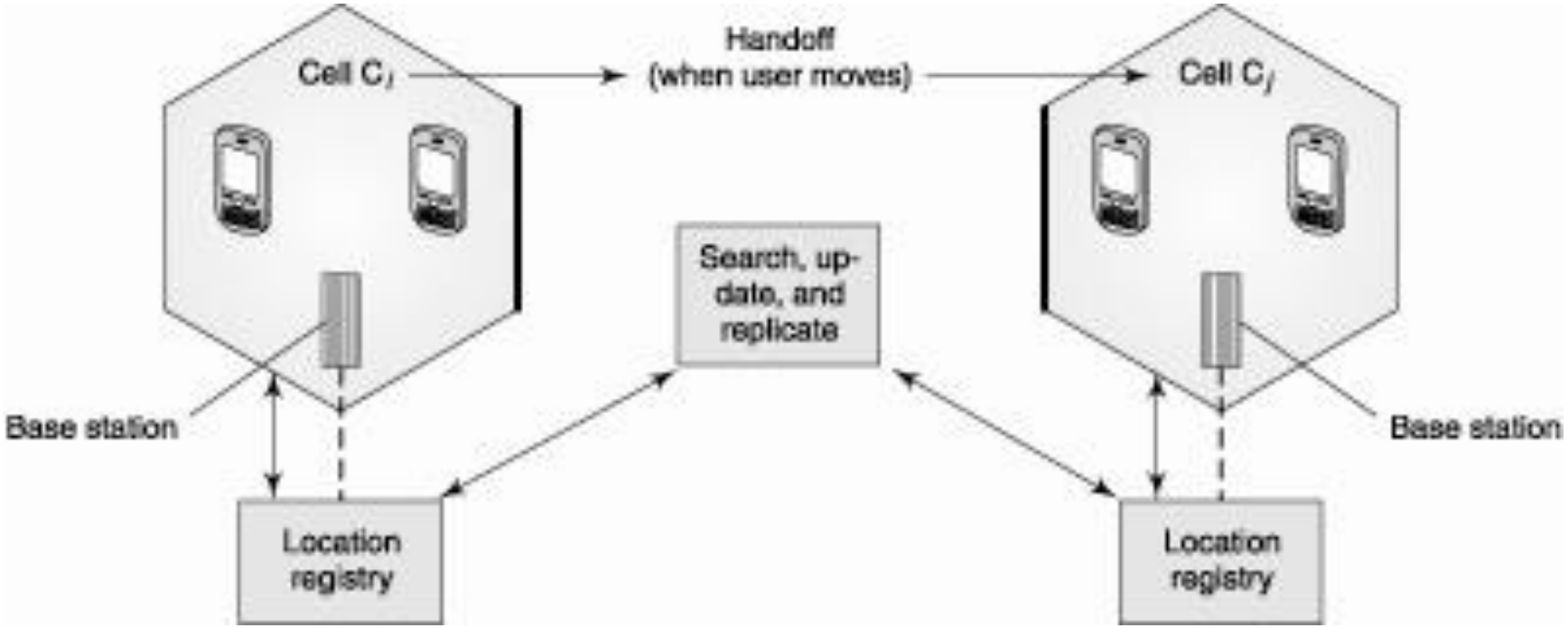
- A base station is responsible to communicate with hosts in its cell
- Mobile hosts can change cells while communicating
- Hand-off occurs when a mobile host starts communicating via a new base station

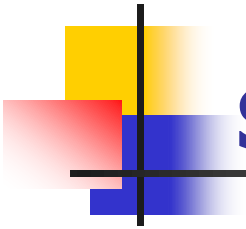


# Hand-Off Procedure

- Each base station periodically transmits beacon
- Mobile host, on hearing stronger beacon from a new BS, sends it a greeting
  - changes routing tables to make new BS its default gateway
  - sends new BS identity of the old BS
- New BS acknowledges the greeting and begins to route MH's packets

# Hand-Off Procedure





# Subscriber Identity Module (SIM)

- An inserted card provided by the GSM service provider
- Uniquely identifies the user to the service
- Enables the MS to connect to the GSM network
- When the MS connects to the GSM subsystems, the SIM saves
  - Temporary Mobile Subscriber Identity (TMSI)
  - Location Area Identification (LAI)
- Information which does not change when the MS moves into another location
  - International Mobile Subscriber Identity (IMSI)
  - Card serial number and type

# Home Location Register (HLR)

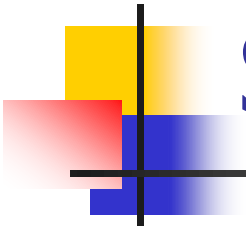
- Has the databases
- Stores all the relevant subscriber  data including
  - mobile subscriber ISDN number details of subscription
  - permissions such as call forwarding, roaming, etc.
  - subscriber's ISMI
  - user's location area,
  - user's subscriber's ISMI
  - user's location area
  - user's current VLR
  - MSC status

# Home Location Register (HLR)

- Each mobile user has only one HLR record worldwide, which is updated constantly on a real-time basis
- Each MS must register at a specific HLR of a specific MSC
- Each HLR is associated to an MSC so that when an MS registered at a certain HLR moves to another location area (LA), serviced by another MSC, the user's home MSC update the user's current VLR

# Visitor Location Register (VLR)

- Each MSC has a dynamic real-time database that stores
- both permanent and temporary subscriber Data
- required for communication between the MSs in the coverage area of the MSC associated with that VLR.



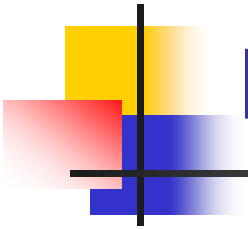
# Switching Techniques

- Circuit Switching
  - A connection first sets up and then the entire data transmits through the path that has been set up
- Packet switching
  - Packets of data at any given instant can take multiple (time slots or channels or paths or routes)
  - Depending on the idle slots at that instant
  - Receiver assembles the packets into the original sequence in the data

# GPRS

**General packet radio service**

- A speed enhanced data transmission service designed for GSM systems
- By packetizing data and simultaneous transmission of packets over different Channels
- Uses the unused slots and channels in TDMA mode of a GSM network for packetized transmission from a mobile station
- Facilitates Internet Access at medium speed



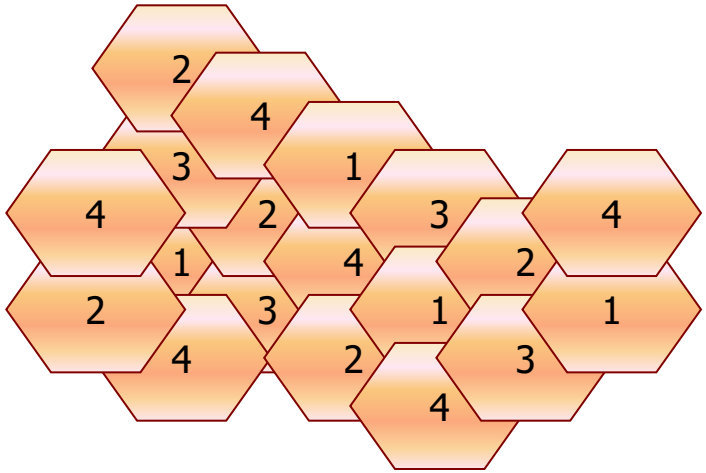
# EDGE

- **EDGE Enhanced Data Rates for Global Evolution**
  - EDGE is add-on to GPRS
  - Uses 8-PSK modulation
  - Offer data rates of 384kbps, theoretically up to 473.6kbps
- New handsets / terminal equipment; additional hardware in the BTS, Core network and the rest remains the same
- EDGE access develops to connect to 3G core

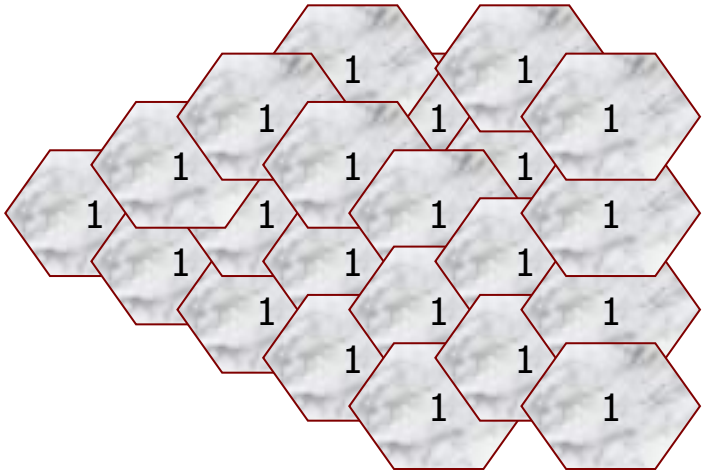
# Code Division Multiple Access-CDMA

- A Chip sequence assigned to each mobile device
- Each user uses the same frequency all the time, but mixed with different distinguishing code patterns

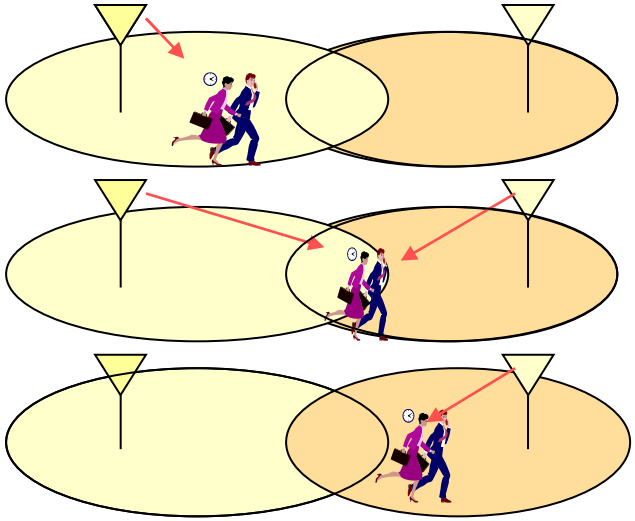
**GSM: N=4**
**Frequency reuse**

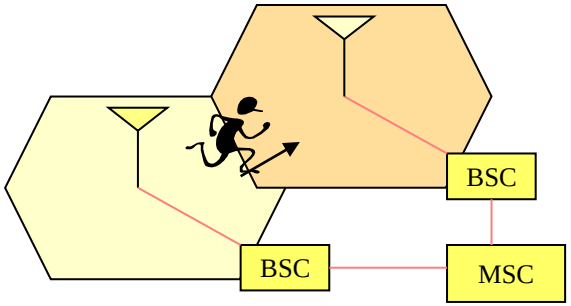



**CDMA: N=1**
**Frequency reuse**

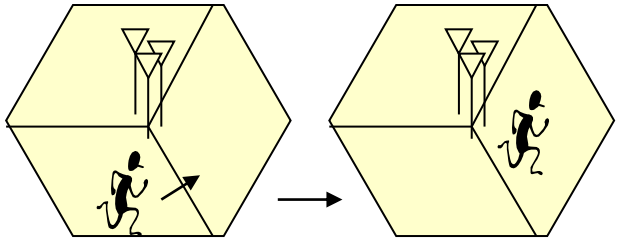


# Soft and Softer Handoff





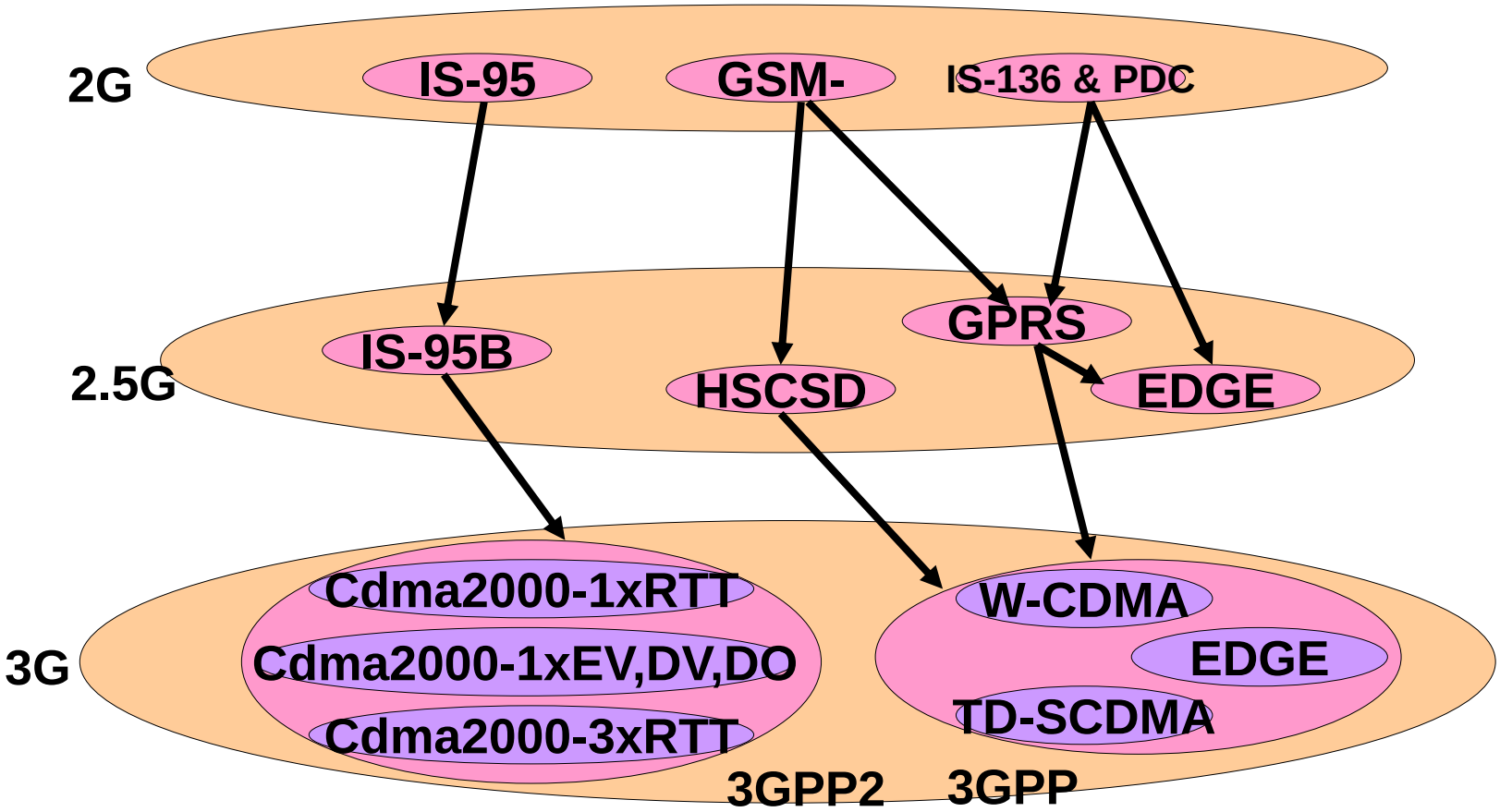

Soft handoff: adjacent cells of the same frequency.



Softer handoff: same BS & frequency, between different sectors.

**Connection first and disconnection afterwards in handoff.**
**High quality of service and effective reduction of call drops.**

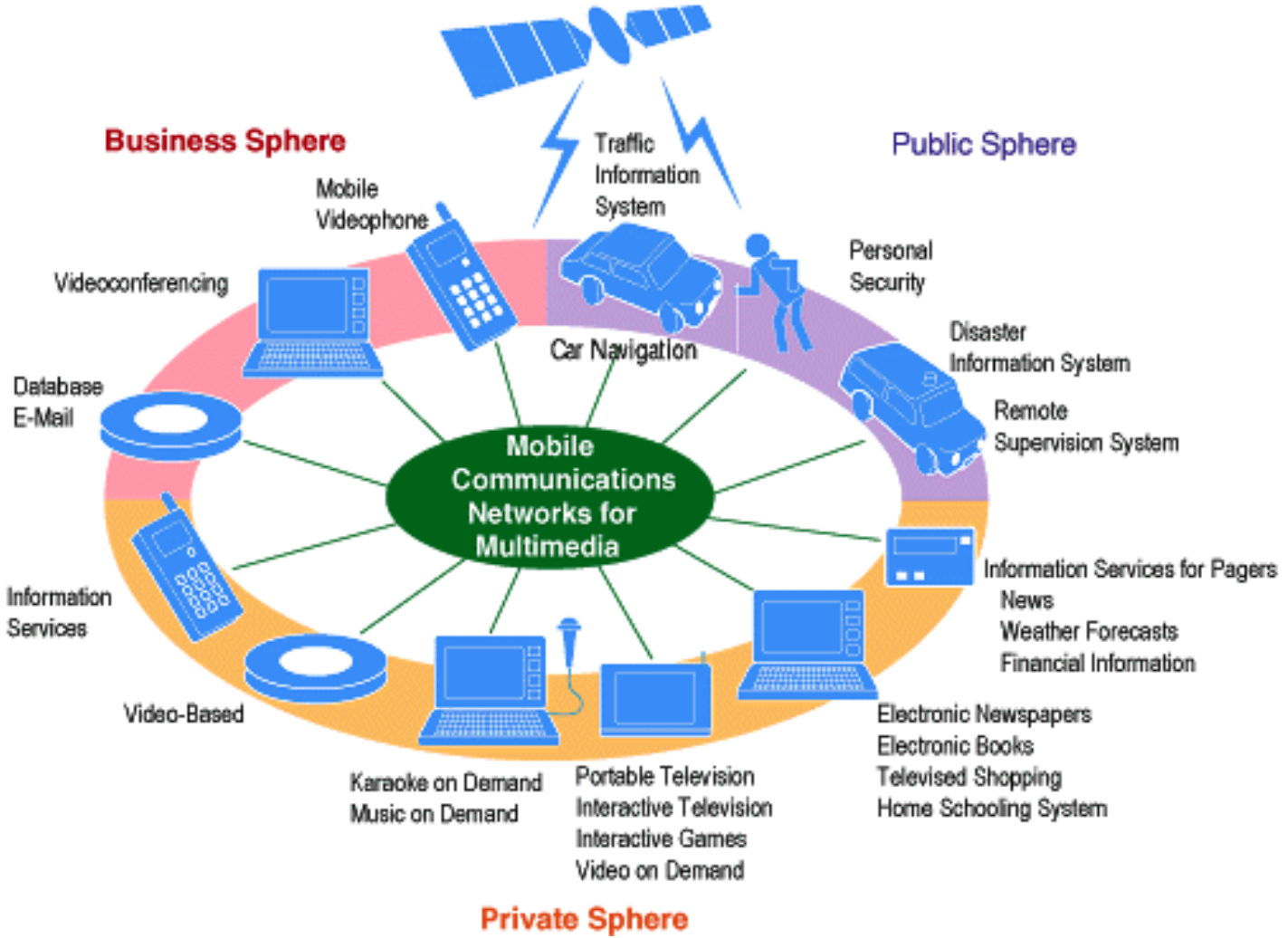
# Upgrade Path



# 3G

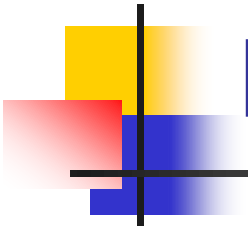
- 3G Standard is created by ITU-T and is called as **IMT-2000.**
- The aim of IMT-2000 is to harmonize worldwide 3G systems to provide **Global Roaming.**
- Higher bandwidth enables a range of new applications!!
- **For the consumer**
  - Video streaming, TV broadcast
  - Video calls, video clips – news, music, sports
  - Enhanced gaming, chat, location services…
- **For business**
  - High speed teleworking / VPN access
  - Sales force automation
  - Video conferencing
  - Real-time financial information

# A world of mobile multimedia



# 4G: Anytime, Anywhere Connection

- Also known as ‘Mobile Broadband everywhere’
- ‘MAGIC’
  - Mobile Multimedia Communication
  - Anywhere, Anytime with Anyone
  - Global Mobility Support
  - Integrated Wireless Solution
  - Customized Personal Service
- Transmission at 20 Mbps
- Companies developing 4G technology
  - Cellular phone companies: Alcatel, Nortel, Motorola,
  - IT Companies: Hughes.HP.LG Electronics



# Limitations of Mobile Environment

- Limitations of the Wireless Network
  - heterogeneity of fragmented networks
  - frequent disconnections
  - limited communication bandwidth
- Limitations Imposed by Mobility
  - lack of mobility awareness by system/applications
- Limitations of the Mobile Computer
  - short battery lifetime
  - limited capacities